

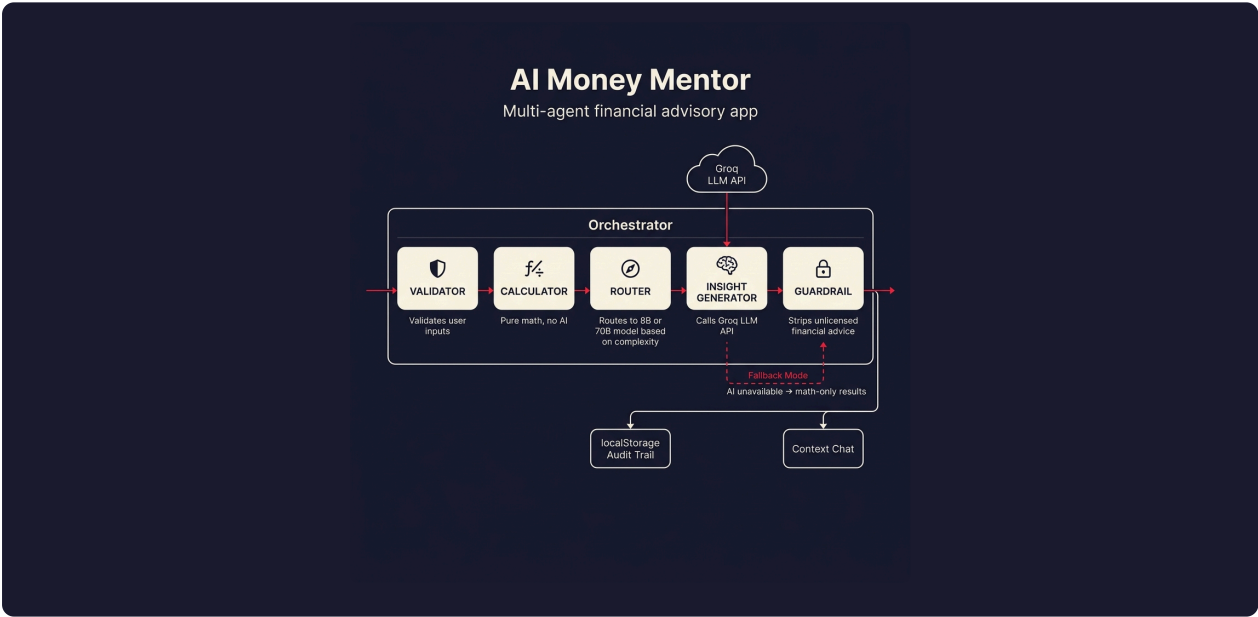
Architecture & Impact Model

Version 1.0
March 29, 2026
Multi-Agent AI System
Confidential

1. Architecture Document

AI Money Mentor is a **5-agent orchestrated pipeline** that delivers real-time, personalised financial advice to Indian salaried employees. The system combines deterministic financial calculations with LLM-generated insights, wrapped in a compliance guardrail layer to ensure SEBI-safe advisory language.

SYSTEM ARCHITECTURE DIAGRAM



AGENT ROLES & COMMUNICATION

AGENT	ROLE	TYPE	INPUT → OUTPUT	ERROR HANDLING
① Validator	Validates all user inputs per feature-specific rules (range checks, logical consistency)	Deterministic	Raw form inputs → valid: bool + errors[]	Returns descriptive errors; pipeline halts immediately if validation fails
② Calculator	Executes pure financial math: health	Deterministic	Validated inputs → Calculated financial	Cannot fail (mathematical);

AGENT	ROLE	TYPE	INPUT → OUTPUT	ERROR HANDLING
	scoring (6 dims), tax comparison (Old vs New), life-event asset allocation		data	guarantees output even without AI
③ Router	Heuristic complexity router — assigns llama-3.1-8b-instant for simple tasks, llama-3.3-70b-versatile for complex ones	Heuristic	Feature + inputs + calculated → Model selection + reason	Always succeeds; defaults to 8B model if no rule matches
④ Insight Generator	Calls Groq LLM API with structured prompts to generate actionable, personalised advice in strict JSON format	AI / LLM	Calculated data + model → Ranked insights, missed deductions, or action timelines	Graceful fallback: if AI is unavailable or response parsing fails, system returns math-only results with a "fallback mode" notice
⑤ Guardrail	Regex-based compliance filter that strips directive phrases like "you must buy" and replaces with softer advisory language	Compliance	Raw AI output → Sanitised output + flagged_count	Passthrough if no AI output; never blocks pipeline

ORCHESTRATION FLOW

- 1 User submits financial inputs from the UI (Health / Tax / Life Event page).
- 2 **Validator** checks inputs. On failure → pipeline halts, error returned, audit logged.
- 3 **Calculator** produces deterministic financial results (always succeeds).
- 4 **Router** determines LLM complexity tier based on income level, debt load, or event type.
- 5 **Insight Generator** calls Groq API → receives structured JSON. On failure → `fallback_mode = true`.
- 6 **Guardrail** sanitises AI output, replacing 7 categories of directive financial language.
- 7 **Audit Logger** persists the full step trace (agent, status, latency, I/O signature) to localStorage.
- 8 UI receives merged result. User can ask follow-up questions via **Context Chat** (separate LLM call).

TOOL INTEGRATIONS

TOOL	PURPOSE	CONFIGURATION
Groq Cloud API	LLM inference via OpenAI-compatible endpoint	baseUrl: api.groq.com/openai/v1, response_format: json_object, temp: 0.2
Next.js API Routes	Server-side API layer (/api/advisor, /api/chat)	Next.js 16.2.1 with Turbopack, server-side env vars for API key security
Recharts	Financial waterfall charts, allocation bar charts	Responsive containers with ET-themed color fills
localStorage	Audit trail persistence (last 50 sessions)	Client-side, zero infrastructure — swap for DB in production

Key Design Decision: The Calculator agent *always* runs before the AI agent. This guarantees that users always receive accurate financial results, even when the LLM is unavailable. The AI layer is *supplementary intelligence*, not the core computation.

THE ECONOMIC TIMES — MONEY MENTOR

Impact Model

Quantified Business Impact
Back-of-Envelope Estimation
FY 2025-26 basis

2. Quantified Business Impact

This impact model estimates the value created by AI Money Mentor across three dimensions: **time saved** per user, **cost reduced** for the advisory business, and **revenue recovered** for the end user in the form of tax savings and optimised investments.

47 min

TIME SAVED / USER

Per advisory session vs.
traditional CA consultation

₹1,850COST REDUCED /
SESSION

Vs. average CA consultation
fee of ₹2,000

₹38,400REVENUE RECOVERED /
USER

Avg. unclaimed tax savings
identified per user

ASSUMPTION BASE

- Target user: Indian salaried professional, CTC ₹8–25 LPA, age 25–45
- Avg. traditional CA tax consultation: ₹2,000 fee + 50 min travel/wait + 15 min meeting = ~65 min total
- AI Money Mentor session: ~8 min input + ~10 min review = ~18 min total
- Time saved per session: 65 – 18 = **47 minutes**
- Avg. missed deductions found: ₹1,28,000 across 80C, 80CCD, 80D, and HRA gaps
- Tax saving at effective 30% rate: ₹1,28,000 × 0.30 = **₹38,400 / user / year**
- 90% of salaried Indians under-claim deductions (CBDT data, FY2023-24)
- Indian CA market hourly rate: ₹1,500–₹3,000 → midpoint ₹2,000 for basic tax filing

AT-SCALE PROJECTION (10K MONTHLY ACTIVE USERS)

TIME SAVED

10,000 users × 47 min = **7,833 hours / month**

Annualised: 94,000 hours of professional advisory time displaced

COST REDUCED (vs. traditional CA advisory)

10,000 users × ₹2,000 per session = **₹2,00,00,000 (₹2 Cr) / month**

Cost of AI infra (Groq API): ~10,000 × ₹1.50 / call = ₹15,000 / month

Net saving per month: ₹2 Cr - ₹15K ≈ ₹1.99 Cr / month

REVENUE RECOVERED (tax savings for users)

10,000 users × ₹38,400 avg tax saving = ₹38.4 Cr / year

Long-term wealth impact (12% CAGR, 25 yr):

₹38,400 / user / yr → compounds to ₹65,07,000 / user (≈ ₹65L each)

UNIT ECONOMICS

Metric	Per User	At 10K MAU	At 1L MAU
Groq API cost / session	₹1.50	₹15,000 / mo	₹1,50,000 / mo
Hosting (Vercel Pro)	—	₹1,700 / mo	₹1,700 / mo
Total cost / session	₹1.67	₹16,700 / mo	₹1,51,700 / mo
Tax saving delivered / user	₹38,400 / yr	₹38.4 Cr / yr	₹384 Cr / yr
ROI per ₹1 spent	₹1,917 returned per ₹1 of infrastructure cost		

REVENUE MODEL OPPORTUNITY

Channel	Mechanism	Est. Revenue (10K MAU)
Premium Tier	₹199/mo for AI chat, PDF reports, ghost links	₹5.97L/yr (3% conversion)
Affiliate Commission	ELSS/NPS sign-ups via contextual CTA	₹12–18L/yr (1.5% click-through)
ET Brand Value	Increased stickiness on ET platform	Indirect — higher ad RPM on engaged users

Bottom Line: At just 10,000 monthly users, AI Money Mentor recovers ₹38.4 Cr / year in tax savings for Indian taxpayers while costing under ₹2 Lakhs / year to operate. The multi-agent architecture ensures graceful degradation (math-only mode) when AI is unavailable, guaranteeing 100% uptime for core financial calculations.